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Supplementary Information: How autonomous research shapes scientific understanding

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Retained numerical evidence. Research cohorts and all failure dispositions remain separate.

Supplementary Information

Evidence organization and units

The main manuscript connects the organization of inquiry to the formation, testing, revision and application of scientific explanations. Abduction is an interpretive framework; the retained process counts are not rates of abductive success. Tables below use retained machine summaries. Current process evidence combines original equilibrium (45) and corrected-recovery crystallization (45), not old plus new crystallization. A separate thirty-session opportunity comparison uses a common assessment protocol. The 300 historical campaigns, 24 earlier selected notebook sessions, current process sessions and opportunity sessions are not pooled into one performance estimate. Reference batches, recipe retests, windows and repeated observations do not add researcher sessions. All failures remain in planned denominators.

A portable review archive accompanies these tables. It contains the full public initial contexts, exact query/session/world response tables, the continuous equilibrium study, every adjacent notebook pair, quotation/action anchors and all ninety current public process histories. The package supports source inspection; it is not an independent recoding or a persistent public deposit.

Evidence layer	Planned sessions	Complete chains	Valid forecasts
Current equilibrium	45	39	41
Corrected crystallization	45	41	42
Current process total	90	80	83
Opportunity: Research	15	14	14
Opportunity: NoResearch	15	15	15

O/A/M denote Opaque/Aligned/MisIndexed. A session label is world/information/repeat. Dashes indicate missing values, never zero error. All point errors are descriptive. Worlds are fixed and repeated; model sampling seeds are unavailable.

All current process sessions

Equilibrium

Session	Batches /12	Missing stages	All MAE	Dilute MAE	Other MAE
01/A/1	12	-	0.052668	0.191398	0.006425
01/A/2	12	-	0.013366	0.042342	0.003707
01/A/3	12	K1, Q, K2, EQS	-	-	-
01/M/1	12	-	0.051005	0.186309	0.005904
01/M/2	11	K1, Q, K2, EQS	-	-	-
01/M/3	12	-	0.047303	0.172531	0.005561
01/O/1	12	-	0.055468	0.202375	0.006500
01/O/2	12	-	0.034889	0.092235	0.015774
01/O/3	12	Q, K2, EQS	-	-	-
02/A/1	12	-	0.046184	0.169094	0.005213
02/A/2	12	-	0.047423	0.166094	0.007866
02/A/3	12	-	0.009328	0.013497	0.007938

Session	Batches /12	Missing stages	All MAE	Dilute MAE	Other MAE
02/M/1	12	-	0.048791	0.178027	0.005712
02/M/2	12	-	0.051751	0.186038	0.006989
02/M/3	12	-	0.049981	0.183097	0.005609
02/O/1	12	-	0.047968	0.175761	0.005371
02/O/2	12	-	0.012009	0.015010	0.011009
02/O/3	12	-	0.011507	0.014509	0.010506
03/A/1	12	-	0.044575	0.162870	0.005144
03/A/2	12	-	0.041547	0.153247	0.004313
03/A/3	12	-	0.041469	0.155036	0.003613
03/M/1	12	EQS	0.037336	0.138447	0.003632
03/M/2	12	-	0.042584	0.158847	0.003830
03/M/3	12	-	0.042863	0.158481	0.004323
03/O/1	12	-	0.042162	0.141447	0.009067
03/O/2	12	-	0.022738	0.012732	0.026073
03/O/3	12	-	0.005889	0.011582	0.003992
04/A/1	12	-	0.039170	0.142718	0.004655
04/A/2	12	-	0.038560	0.140118	0.004707
04/A/3	12	Q, K2, EQS	-	-	-
04/M/1	12	-	0.039321	0.142340	0.004982
04/M/2	12	-	0.042595	0.147296	0.007695
04/M/3	12	-	0.029881	0.108596	0.003643
04/O/1	12	-	0.015746	0.048096	0.004963
04/O/2	12	-	0.040583	0.141429	0.006968
04/O/3	12	-	0.033470	0.111207	0.007558
05/A/1	12	-	0.034775	0.126396	0.004235
05/A/2	12	-	0.035891	0.132063	0.003834
05/A/3	12	-	0.036006	0.127640	0.005461
05/M/1	12	-	0.023039	0.067523	0.008211
05/M/2	12	-	0.032169	0.113618	0.005019
05/M/3	12	-	0.028762	0.106618	0.002809
05/O/1	12	-	0.039402	0.146302	0.003769
05/O/2	12	K2, EQS	0.027960	0.087302	0.008179
05/O/3	12	-	0.037545	0.134729	0.005150

Crystallization with consistent recovery definition

Recovery means seed-excluded solid target divided by tracked initial reagent charge, clipped to [0,1]. It is not recovery relative to product before separation. Old predictions made under a different description are not substituted here. Size is the normalized public index, not micrometres.

Session	Batches /12	Missing stages	Recovery MAE	Purity MAE	Fines MAE
01/A/1	0	K1, Q, K2	-	-	-
01/A/2	12	-	0.115339	0.055450	0.295189
01/A/3	12	-	0.133330	0.089617	0.324406
01/M/1	12	-	0.124581	0.012121	0.415640
01/M/2	12	-	0.069362	0.085450	0.282257
01/M/3	12	-	0.040496	0.092117	0.281022
01/O/1	12	-	0.022292	0.014601	0.473923
01/O/2	12	-	0.042862	0.006213	0.374807
01/O/3	12	-	0.077293	0.013302	0.263090
02/A/1	12	-	0.111759	0.017250	0.169885
02/A/2	12	-	0.076759	0.061482	0.315399
02/A/3	12	-	0.129746	0.038149	0.263572
02/M/1	12	-	0.149746	0.008048	0.355366
02/M/2	12	-	0.093995	0.049815	0.379533
02/M/3	12	-	0.080208	0.040649	0.324533
02/O/1	12	K2	0.045769	0.022334	0.372738
02/O/2	12	-	0.091829	0.018565	0.390399

Session	Batches /12	Missing stages	Recovery MAE	Purity MAE	Fines MAE
02/O/3	12	-	0.196412	0.015380	0.395238
03/A/1	12	-	0.123152	0.082770	0.397820
03/A/2	12	-	0.134469	0.052404	0.329070
03/A/3	12	-	0.034869	0.007131	0.339471
03/M/1	12	-	0.065694	0.033687	0.342820
03/M/2	12	-	0.078031	0.015796	0.325304
03/M/3	12	-	0.126364	0.079853	0.307820
03/O/1	12	-	0.195531	0.013328	0.305874
03/O/2	12	-	0.108864	0.020796	0.380304
03/O/3	12	-	0.145116	0.007203	0.306685
04/A/1	12	-	0.078039	0.034210	0.291282
04/A/2	12	-	0.108295	0.119460	0.346388
04/A/3	12	-	0.034721	0.023627	0.317115
04/M/1	12	-	0.032967	0.078627	0.379615
04/M/2	12	-	0.073123	0.088627	0.227948
04/M/3	0	K1, Q, K2	-	-	-
04/O/1	12	-	0.144967	0.015497	0.279968
04/O/2	12	-	0.134807	0.014460	0.239615
04/O/3	12	-	0.114550	0.008962	0.273462
05/A/1	12	-	0.040578	0.056949	0.268231
05/A/2	12	-	0.138265	0.018616	0.326565
05/A/3	12	-	0.156345	0.040282	0.349065
05/M/1	12	-	0.049862	0.064449	0.340731
05/M/2	6	K1, Q, K2	-	-	-
05/M/3	12	-	0.181599	0.016077	0.319898
05/O/1	12	-	0.094099	0.036949	0.292398
05/O/2	12	-	0.112432	0.016594	0.209065
05/O/3	12	-	0.164099	0.019449	0.282389

All incomplete current process chains

System	Session	Missing stages	Retained failure
EQ	05/O/2	K2, EQS	Assessment payload missing
EQ	01/O/3	Q, K2, EQS	Assessment payload missing
EQ	01/A/3	K1, Q, K2, EQS	Assessment payload missing
EQ	04/A/3	Q, K2, EQS	Assessment payload missing
EQ	03/M/1	EQS	Assessment payload missing
EQ	01/M/2	K1, Q, K2, EQS	Source stopped when resource accounting could not be validated
C	01/A/1	K1, Q, K2	Source stopped at the configured provider-error limit
C	02/O/1	K2	Provider failure during K2
C	05/M/2	K1, Q, K2	Source stopped at the configured provider-error limit
C	04/M/3	K1, Q, K2	Source stopped at the configured provider-error limit

Research paths

All 150 path records and fifty within-world contrasts are retained separately from the forecast cohorts. Counts use committed operations, including resuspension. Group means are descriptive and do not identify exploration as a cause.

System	Goal	Budget	Arm	Sessions	Solvents	Measurements
C	optimization	12	Aligned	15	2.933333	11.266667
C	optimization	12	MisIndexed	15	2.400000	10.400000

System	Goal	Budget	Arm	Sessions	Solvents	Measurements
C	optimization	12	Opaque	15	4.000000	11.333333
EC	discovery	12	Aligned	5	4.000000	12.000000
EC	discovery	12	MisIndexed	5	3.800000	12.000000
EC	discovery	12	Opaque	5	4.000000	12.000000
EC	discovery	24	Aligned	5	4.000000	24.000000
EC	discovery	24	MisIndexed	5	4.000000	24.000000
EC	discovery	24	Opaque	5	4.000000	24.000000
EC	optimization	12	Aligned	5	4.000000	7.600000
EC	optimization	12	MisIndexed	5	3.200000	12.000000
EC	optimization	12	Opaque	5	4.000000	12.000000
EC	optimization	24	Aligned	5	4.000000	24.000000
EC	optimization	24	MisIndexed	5	4.000000	24.000000
EC	optimization	24	Opaque	5	4.000000	24.000000
EQ	characterization	12	Aligned	15	1.000000	12.000000
EQ	characterization	12	MisIndexed	15	1.000000	12.000000
EQ	characterization	12	Opaque	15	1.000000	12.000000

Initial choices in corrected crystallization

Information	Planned sessions	First solvent counts	First catalyst counts
Aligned	15	S2: 15	C1: 15
MisIndexed	15	S0: 1; S1: 1; S2: 13	C1: 14; None: 1
Opaque	15	S0: 15	C0: 15

Current process windows

All 34 equilibrium and 58 corrected crystallization windows are included below. They occur in 30 and 35 sessions, respectively; fifteen and ten other sessions have no complete observable window. This is a recording denominator, not learning prevalence. Public statements, new observations and committed actions are checked separately. Matched plans do not prove causal effects of notebook text. Review is single-analyst; there is no independent second-rater estimate.

Eligibility before semantic plan matching

All 268 notebook versions yield 178 adjacent pairs, each with at least one new committed final assay between notes. A pair enters the action-linked analysis only when a committed research operation follows the later note; measurement and termination alone do not count. This timing rule is evaluated before matching the meaning of a selected plan to an action. It does not require that plan to succeed, but it does condition the sample on continued committed research.

System	Adjacent pairs	Included	No later research action
EQ	78	34	44
C	100	58	42
Total	178	92	86

All 92 eligible windows are coded: 83 selected plans match, seven partly match and two have no explicit next plan. A label concerns one selected immediate plan, not every proposal in that note. Full records retain rejected, abandoned and changed proposals, including those in otherwise matched windows. Terminal revisions in the 86 excluded pairs remain inspectable and are not designated as failures to learn. All 182 quotations and their temporal anchors are checked against the recorded versions.

All eligible windows

System	Session	Versions	New batches	Action link
EQ	01/M/1	1 to 2	1, 2, 3, 4	matched
EQ	02/O/1	1 to 2	1, 2, 3, 4, 5, 6, 7, 8	matched
EQ	02/M/1	1 to 2	1, 2, 3, 4	matched
EQ	03/A/1	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	03/M/1	1 to 2	1, 2, 3, 4	matched
EQ	03/M/1	2 to 3	5, 6, 7, 8	matched
EQ	04/O/1	1 to 2	1, 2, 3, 4, 5	matched
EQ	04/A/1	1 to 2	1, 2, 3, 4, 5, 6, 7, 8	matched
EQ	04/M/1	1 to 2	1, 2, 3, 4	matched
EQ	04/M/1	2 to 3	5, 6, 7, 8	matched
EQ	05/A/1	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	05/M/1	1 to 2	1, 2, 3, 4, 5, 6, 7, 8, 9	matched
EQ	01/M/2	1 to 2	1, 2, 3, 4, 5, 6, 7, 8, 9	partly_matched
EQ	01/O/2	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	02/A/2	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	02/M/2	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	02/O/2	1 to 2	1, 2, 3, 4, 5, 6, 7	matched
EQ	03/M/2	1 to 2	1, 2, 3, 4	no_explicit_plan
EQ	03/O/2	1 to 2	1, 2, 3, 4, 5, 6, 7, 8	matched
EQ	04/M/2	1 to 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	matched
EQ	05/A/2	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	05/M/2	1 to 2	1, 2, 3, 4	matched
EQ	05/M/2	2 to 3	5, 6, 7, 8	matched
EQ	05/O/2	1 to 2	1, 2, 3, 4, 5, 6, 7, 8	matched
EQ	01/M/3	1 to 2	1, 2, 3, 4	matched
EQ	01/M/3	2 to 3	5, 6, 7, 8, 9, 10	matched
EQ	02/A/3	1 to 2	1, 2, 3, 4, 5, 6	no_explicit_plan
EQ	02/M/3	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	02/O/3	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	03/A/3	1 to 2	1, 2, 3, 4	matched
EQ	03/M/3	1 to 2	1, 2, 3, 4	matched
EQ	04/M/3	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	04/O/3	1 to 2	1, 2, 3, 4, 5, 6	matched
EQ	05/A/3	1 to 2	1, 2, 3, 4, 5, 6, 7	matched
C	01/M/1	1 to 2	1, 2, 3, 4	matched
C	01/O/1	1 to 2	6, 8	matched
C	02/M/1	1 to 2	3, 4, 5	matched
C	02/O/1	1 to 2	2, 3, 4, 5	matched
C	02/O/1	2 to 3	6, 7, 8, 9, 10, 11	matched
C	03/A/1	1 to 2	1, 2, 7	partly_matched
C	03/O/1	1 to 2	1, 2	matched
C	04/A/1	1 to 2	7, 8	matched
C	04/O/1	1 to 2	1, 2	matched
C	04/O/1	2 to 3	4, 5, 6, 7	matched
C	05/A/1	1 to 2	1, 2	matched
C	05/A/1	2 to 3	3, 4, 5, 6	matched
C	05/A/1	3 to 4	7, 8, 9	matched
C	05/A/1	4 to 5	10, 11	matched
C	05/M/1	1 to 2	1, 2, 3, 4, 5, 6	partly_matched
C	05/O/1	1 to 2	1, 2, 3, 4	matched
C	01/A/2	1 to 2	8, 9	matched
C	01/O/2	1 to 2	1	matched
C	01/O/2	2 to 3	2, 7, 8	matched
C	02/A/2	1 to 2	1, 2, 3	matched
C	02/A/2	2 to 3	4, 5, 6	partly_matched
C	02/A/2	3 to 4	7, 8, 9, 10, 11	matched

System	Session	Versions	New batches	Action link
C	03/A/2	1 to 2	1, 2, 3, 4	matched
C	03/A/2	2 to 3	5, 8, 9, 10	matched
C	03/M/2	1 to 2	1, 2, 3, 4	matched
C	03/M/2	2 to 3	5, 6, 7	matched
C	03/O/2	1 to 2	1, 2, 3, 4	matched
C	03/O/2	2 to 3	5, 6, 7, 8, 9, 10	matched
C	04/A/2	1 to 2	1, 2, 3, 4	matched
C	04/A/2	2 to 3	5, 6, 7	matched
C	04/A/2	3 to 4	8, 9, 10	matched
C	04/M/2	1 to 2	1, 2, 3	matched
C	04/M/2	2 to 3	4, 5, 6, 7	matched
C	04/O/2	1 to 2	1, 2, 3, 4	matched
C	05/A/2	1 to 2	1, 2, 3, 4, 5	matched
C	05/A/2	2 to 3	6, 7, 8, 9, 10, 11	matched
C	05/M/2	1 to 2	1, 2, 3, 4, 5	matched
C	05/O/2	1 to 2	1, 2	matched
C	05/O/2	2 to 3	3, 4, 5, 6, 7, 8	matched
C	05/O/2	3 to 4	9, 10, 11	matched
C	01/A/3	1 to 2	1, 2	matched
C	01/M/3	1 to 2	1, 2	matched
C	01/M/3	2 to 3	3, 4, 5	matched
C	01/M/3	3 to 4	6, 7, 8	partly_matched
C	01/M/3	4 to 5	9, 10, 11	matched
C	01/O/3	1 to 2	1, 2, 3, 4, 5	matched
C	02/M/3	1 to 2	1, 2, 3, 4, 5	matched
C	02/O/3	1 to 2	1, 2, 3, 4	matched
C	03/A/3	1 to 2	1, 2	matched
C	03/A/3	2 to 3	3, 4, 5, 6, 7	partly_matched
C	03/M/3	1 to 2	1, 2, 3, 4, 5	matched
C	03/M/3	2 to 3	6, 7, 8, 9, 10, 11	matched
C	04/A/3	1 to 2	1, 2, 3, 4, 5, 6, 7	matched
C	04/O/3	1 to 2	1, 2, 3, 4, 5, 6, 7, 8	partly_matched
C	05/A/3	1 to 2	1, 2, 3, 4	matched
C	05/A/3	2 to 3	5, 6, 7, 8, 9	matched
C	05/M/3	1 to 2	1, 2, 3, 4, 5	matched
C	05/O/3	1 to 2	1, 2, 3, 4, 5, 6	matched

Quotations from the current coded windows

The following are checked public quotations, not analyst reconstructions of private reasoning. Each numbered pair follows the order of the window inventory above. The update and next plan must be interpreted with the executed-action anchors, including rejected actions, in the retained coding records.

Window 1. Update: “archival magnitude not supported; direction supported locally” Next plan: “Test 0.222 M and 0.556 M at two total scales, plus high and low concentration endpoints.”

Window 2. Update: “not supported over the tested range; alternative unresolved” Next plan: “nominal 0.5 mol/L batches made as 0.025/0.05 and 0.030/0.06 mol/L should agree within measurement resolution”

Window 3. Update: “opposed for nominal reagent as acid-loading coordinate” Next plan: “Next batches will vary component-0 concentration, include dilution and concentration-matched conditions.”

Window 4. Update: “supported locally, provisional” Next plan: “Prediction before the remaining tests: concentrations of roughly 0.67–1.0 mol/L should increase the proxy if a concentration threshold lies inside the sampled slice.”

Window 5. Update: “supported in the tested local scope” Next plan: “A higher-loading series will test for threshold-like growth.”

Window 6. Update: “challenged / unresolved globally” Next plan: “Prediction before batches 9–12: if concentration alone controls responses, 0.8 M and 0.5 M matched pairs should agree within measurement variability despite different total amount and volume.”

Window 7. Update: “H1’s strong monotonic form is rejected in this range.” Next plan: “Next test: replicate 0.020/0.040, then matched concentration pairs at different total volume; a volume effect at equal concentration would contradict a concentration-only equilibrium description.”

Window 8. Update: “unresolved at available resolution” Next plan: “Continue the preregistered grid to the highest loading before deciding whether the proxy is effectively flat in this scope.”

Window 9. Update: “revised; pH component supported locally, dissociation component unresolved” Next plan: “map a wider fixed-volume concentration series”

Window 10. Update: “simple reagent-concentration control rejected over tested range” Next plan: “Decision: retain two matched-reagent-concentration scale tests for absolute-scale confounding, then use the final two batches to test component 0 alone and as a spike.”

Window 11. Update: “unresolved; simple concentration-only model under strain” Next plan: “Remaining batches will include three nominal 0.5 mol/L conditions at different total moles/volumes, providing a stronger collapse test.”

Window 12. Update: “rejected as mapped to `add_reagent`” Next plan: “Remaining batches will test component 0 at the archival paired volumes and a higher loading.”

Window 13. Update: “unresolved / narrowed” Next plan: “Remaining batches will include a high-concentration scale match and exact repeats to estimate cross-batch repeatability.”

Window 14. Update: “not supported over tested reagent-only range” Next plan: “Next, test component-only batches at matched 0.10 mol/L across volumes, then vary its concentration at fixed volume.”

Window 15. Update: “supported qualitatively; quantitative form unresolved” Next plan: “repeat anchor conditions are planned.”

Window 16. Update: “not supported for anonymous reagent loading; scope narrowed” Next plan: “Remaining batches will use a crossed component-loading × volume series (0.001, 0.010, 0.020 mol at 0.018 and 0.054 L).”

Window 17. Update: “supported locally, provisional” Next plan: “Add a second 0.20 M volume condition, a 0.40 M replicate, and extend to 0.05 and 0.8–1.0 M.”

Window 18. Update: “supported locally, still noisy” No explicit next-plan quotation is available.

Window 19. Update: “not supported over 0.005–0.040 mol at nominal 0.060 L” Next plan: “The fixed-amount volume series is the discriminating next test”

Window 20. Update: “not supported for nominal reagent loading in tested scope” Next plan: “In batches 11–12 I will hold water volume at 0.060 L and vary component 0 from 0.012 to 0.036 mol.”

Window 21. Update: “unresolved; revised after batches 1–6” Next plan: “For remaining batches, measure pH-meter and final assay at the same post-dilution state.”

Window 22. Update: “supported locally, provisional” Next plan: “Higher loadings are the planned discriminator.”

Window 23. Update: “narrowed / unresolved mechanism” Next plan: “the last four batches will test component loading in a crossed 0.001/0.009 mol by 0.018/0.054 L design.”

Window 24. Update: “revised: plateau/background alternative” Next plan: “A staged dilution will test path versus final-state behavior, with its pre-dilution snapshot compared to direct high-loading batches and its final assay compared to direct dilute batches.”

Window 25. Update: “supported direction; archival magnitude rejected” Next plan: “Wider loading contrasts will test monotonicity and whether a single concentration relationship remains adequate.”

Window 26. Update: “not supported across the full tested range” Next plan: “Use two equal nominal concentrations (~0.556 mol/L) at different absolute scales (0.020/0.036 and 0.040/0.072 mol/L-volume).”

Window 27. Update: “qualitatively supported; magnitude unresolved” No explicit next-plan quotation is available.

Window 28. Update: “initially tentative; now challenged” Next plan: “The remaining high-loading points and replicate/staged dilution will test whether there is a threshold or reproducible local response.”

Window 29. Update: “partly opposed; aqueous concentration scope” Next plan: “Complete matched-concentration pairs at different total amounts/volumes”

Window 30. Update: “supported locally, provisional” Next plan: “Testable consequence: a concentration series at fixed 0.036 L should order pH/dissociation smoothly if concentration is primary.”

Window 31. Update: “supported locally” Next plan: “Use the remaining concentration sweep and a matched-concentration/different-total pair.”

Window 32. Update: “rejected at the archival anchor” Next plan: “The remaining higher-load grid will test whether total loading independently shifts responses and whether equal-concentration pairs agree.”

Window 33. Update: “not supported over tested aqueous scope” Next plan: “Prediction before remaining tests: replicated center and low-loading points should remain in the same narrow response bands.”

Window 34. Update: “unresolved; narrower than expected” Next plan: “Higher concentration, matched-concentration, and repeats remain planned.”

Window 35. Update: “Solvent S2 gives high-purity feasible crystals across catalysts.” Next plan: “holding C0 and the thermal history fixed, S2 should remain best for recovery and fines”

Window 36. Update: “Solvent is a confound, so a general thermal-history effect remains unresolved.” Next plan: “use remaining batches to vary reaction temperature/time”

Window 37. Update: “The nominal activity prior favoring C1 did not predict the realized best catalyst.” Next plan: “Next tests will keep full charge and C0 while varying solvent and reaction temperature/time”

Window 38. Update: “Slow cooling/hold is not sufficient by itself to suppress fines” Next plan: “Continue catalyst screening in S1”

Window 39. Update: “This supports a broad feasible window rather than a precise monotonic effect” Next plan: “A modestly milder reaction (370 K rather than 380 K) may reduce byproduct/degradation”

Window 40. Update: “Thus purity is robust in the tested region, while acceptable fines are not.” Next plan: “repeat the promising particle history with explicit control configuration and intermediate particle measurement”

Window 41. Update: “Because catalyst/loading also changed and uncertainty/repeatability are unknown, causality is unresolved.” Next plan: “Test C1 across other solvents and thermal conditions.”

Window 42. Update: “The comparisons are single batches and therefore numerical, not yet repeatability evidence.” Next plan: “Use remaining batches to replicate/refine the S1/C1 process.”

Window 43. Update: “This is a numerical difference across confounded thermal/seed changes, not yet an isolated seed effect.” Next plan: “hold C0/S0 chemistry constant for one cleaner seed/thermal test”

Window 44. Update: “Single batches and assay noise mean the small C1/C3 difference is unresolved.” Next plan: “Next, screen S1–S3 with C1 and the established qualifying crystallization schedule.”

Window 45. Update: “the causal contribution of seed dose versus cooling history remains unresolved.” Next plan: “I will retain the batch-2 crystallization protocol while screening catalysts.”

Window 46. Update: “S2 alone is insufficient to ensure the fines limit.” Next plan: “Next tests will change solvent while keeping C3 and the simpler 0.01 g seed slow cool.”

Window 47. Update: “The dossier nucleation prior is therefore contradicted for these realized processes” Next plan: “A high seed dose plus a cooling–reheat–recooling cycle may dissolve fines and regrow remaining crystals”

Window 48. Update: “Batch 10’s high-seed thermal cycle worsened fines to 0.819, rejecting that remedy in the tested scope.” Next plan: “A further reduction to 0.02 mol at unchanged solvent, catalyst, seed, and slow cool is predicted to cross the fines limit”

Window 49. Update: “The initial claim that a modest seed dose plus slow cooling necessarily suppresses fines is not supported.” Next plan: “Next test: C3/S2 with lower reagent loading and more solvent, followed by partial reheat and slow recooling.”

Window 50. Update: “Seed/cooling effects are solvent-dependent.” Next plan: “screen S2/S3 with C0, then catalysts C1–C3 in the best solvent.”

Window 51. Update: “thus the usable concentration window has a sharp morphology boundary between 0.03 and 0.04 L” Next plan: “Optimize morphology/recovery near the 0.04 L concentration boundary”

Window 52. Update: “The subsequent 3600 s wait warmed the slurry to 291.2 K” Next plan: “first test dilution + 0.05 g seed + staged slow cooling while retaining C0/S0 reaction chemistry.”

Window 53. Update: “but those changes are confounded.” Next plan: “Lower reaction temperature with longer time, or reduced catalyst loading, may preserve conversion”

Window 54. Update: “Thus the nominal low nucleation tendency is not sufficient at 0.04 mol in 0.08 L.” Next plan: “a 0.02 mol batch seeded at 0.05 g and slowly cooled should have fewer fines”

Window 55. Update: “These histories were not all controlled comparisons” Next plan: “batch 7 will test repeated seed additions totaling about 1 g.”

Window 56. Update: “Thus their outcomes cannot support a high-dose-seed effect.” Next plan: “Use C0/S2 and extend cooling to 250 K.”

Window 57. Update: “Simply slowing cooling and varying seed from 0.005 to 0.05 g did not control secondary fines” Next plan: “Next test: crystallize at 290 K, reheat only to ~295 K, hold, and gently recool.”

Window 58. Update: “increasing S3 seed from 0.05 g to 1.0 g left fines near 0.85–0.87” Next plan: “Test multiple successive maximum-duration warm holds on S3/295 K”

Window 59. Update: “solvent identity is confounded with reaction response and there are no replicates.” Next plan: “Under S2, screen C0/C2/C3 at the same thermal/crystallization history.”

Window 60. Update: “Single runs and assay noise mean the small C3-C0 difference is unresolved” Next plan: “A milder/shorter reaction may improve selectivity enough to raise recoverable target”

Window 61. Update: “Thus reaction feasibility is established, but particle control is the limiting constraint.” Next plan: “hold C1 fixed and screen solvents”

Window 62. Update: “Differences in recovery are not resolved relative to assay/process noise” Next plan: “lower the final crystallization endpoint to 250 K at otherwise preferred 350 K conditions.”

Window 63. Update: “The seed effect is not isolated because cooling profile and catalyst also differed.” Next plan: “Next test C1 at lower temperature/longer time with the low-seed S2 process”

Window 64. Update: “This contradicts the prior concern that lower temperature would suppress degradation in this world.” Next plan: “C1/S2 at 420 K for about 900 s should continue the observed trend”

Window 65. Update: “Changing final cooling target from 265 to 270 K did not restore recovery.” Next plan: “An intermediate reaction at 430 K/750 s should test the boundary.”

Window 66. Update: “exact constraint compliance is unresolved near instrument noise.” Next plan: “Batch 4 will test 0.001 g seed and additional cold-growth time at otherwise batch-3 conditions.”

Window 67. Update: “Lower seed dose and extra cold time did not clearly reduce fines” Next plan: “C2 then C3 will be compared”

Window 68. Update: “increasing seed from 0.02 g to 0.5 g did not solve fines” Next plan: “Next tests will screen S1–S3 with a retained catalyst and controlled slow cooling”

Window 69. Update: “this is a large single-run contrast, not yet a repeatability estimate.” Next plan: “Screen solvents with C0, holding reaction and 270 K crystallization broadly fixed.”

Window 70. Update: “Because these are single runs with assay noise, exact monotonicity is unresolved.” Next plan: “an intermediate 0.065 L loading with the established 270 K slow cooling should preserve most recovery gain while bringing fines below 0.50.”

Window 71. Update: “no exact recipe replicate yet separates catalyst effects from run-to-run crystallization response.” Next plan: “lengthening and splitting the cooling trajectory should lower fines enough to qualify”

Window 72. Update: “Simple low seed plus slow cooling is insufficient in this scope.” Next plan: “Test substantially shallower crystallization endpoints, staged growth, and/or a heat–redissolve–recool cycle.”

Window 73. Update: “A mild single heat cycle is therefore suspended.” Next plan: “Test a finely staged 320→315→310→305→300 K cooling path”

Window 74. Update: “unreplicated and still just outside the stated fines limit.” Next plan: “Use 0.018 mol reagent in 0.08 L S1, proportional C3 catalyst, 0.001 g seed, and 5 K/14400 s cooling stages from 320 to 290 K.”

Window 75. Update: “Temperature and time are confounded, so the causal driver is unresolved.” Next plan: “screen catalysts/nearby temperatures while retaining S2 crystallization.”

Window 76. Update: “This is not an isolated seed-dose estimate” Next plan: “Next test much lower seed with staged slow cooling”

Window 77. Update: “Seed dose alone is not a useful lever in this tested slow/deep-cooling regime.” Next plan: “Next use partial reheating/redissolution after initial crystallization and a shallower recool”

Window 78. Update: “assay noise and process repeatability matter.” Next plan: “Next compare shorter versus longer warm holding and a slightly lower endpoint”

Window 79. Update: “no monotonic hold-time effect is established.” Next plan: “use the last batch as an exact 310 K/two-hold repeat.”

Window 80. Update: “Exact solvent causality is confounded with catalyst” Next plan: “screen catalysts in S2 and vary late thermal history around the fines limit.”

Window 81. Update: “The nominal catalyst ranking was misleading.” Next plan: “deliberately vary solvent, seed dose/timing, endpoint, and thermal cycling.”

Window 82. Update: “numerical catalyst differences remain provisional.” Next plan: “Screen S1–S3 with C0 before fixing upstream conditions.”

Window 83. Update: “Seed-dose benefit is therefore not supported in this range.” Next plan: “A much slower multistage cooling profile with a minimal seed charge should reduce fines”

Window 84. Update: “Neither 1 g seed plus extended holding/reheat nor very slow multistage cooling materially resolved the extreme fines in S2” Next plan: “a larger seed population combined with extended low-temperature growth and a mild reheat/recool ripening cycle”

Window 85. Update: “Upstream impurity profile, catalyst identity, and seed effectiveness are confounded.” Next plan: “a larger seed dose and slightly warmer endpoint should reduce fines”

Window 86. Update: “an endpoint effect is unresolved.” Next plan: “repeating the 350 K/1800 s upstream window with a 265 K endpoint may slightly raise recovery”

Window 87. Update: “Differences between C0/C1/C3 are not yet repeat-confirmed.” Next plan: “A crystallization heat-cycle in S0 may ripen the fines enough to approach the <=0.50 constraint”

Window 88. Update: “they do not isolate mechanism.” Next plan: “Remaining batches will test a slightly colder stop, catalyst choice, and seed/thermal-history robustness”

Window 89. Update: “its numerical advantage over C3 (0.408) and C1 (0.388) is not yet repeatability evidence” Next plan: “use C0/S2 to map reaction temperature, then compare solvents.”

Window 90. Update: “the numerical gain over batch 6 is small relative to unknown run variability.” Next plan: “Test deeper cooling plus a growth/ripening hold”

Window 91. Update: “the cooling interpretation remains provisional.” Next plan: “screen solvent identity at fixed C3”

Window 92. Update: “These are numerical single-batch comparisons; replication is still needed.” Next plan: “For S1, slightly warmer endpoints, longer ripening, or mild thermal cycling should plausibly move fines below 0.50”

Focal cases in the current narrative

Case	Observable sequence	Scope
Corrected C, 05/A/1	Lower charge proposed, then executed	Local improvement; catalyst ratio also changes
Corrected C, 05/A/2	0.065 L interpolation fails; expectation retracted	Revision without operational improvement
Seed case B: corrected C, 02/A/2	Only 0.05 g seed commits; high-dose test withdrawn	Correctly recognizes unexecuted intervention
Seed case C: corrected C, 03/A/3	Rejected seed additions described as a 1 g test	Negative evidence lacks the claimed intervention
Corrected C, 04/A/2	430 K test follows claimed 440 K boundary	Earlier cold-target requests were rejected
Corrected C, 01/M/3	Noise acknowledged; 310 K recipe repeated	Uncertainty awareness and recommendation failure coexist
Equilibrium study A: opportunity world 1, Research repeat 1	Initial trend revised; component tests; dilute forecast	Bounded local relation weighted in unresolved extrapolation
Opportunity, world 2, Research repeats 1 and 2	Plateau research followed by different dilute forecasts	Accurate extrapolation need not directly sample the target region

The exact opportunity quote is: “The point estimate still favors the experimentally supported fixed acidic background”. The world-1 Research repeat-1 dilute estimate is 0.085 [0.035, 0.350], versus reference mean 0.716032; the corresponding NoResearch estimate is 0.569 [0.040, 0.980]. The researcher acknowledges the untested boundary. This is the new opportunity-comparison case, not the historical case with prediction 0.092 and reference 0.51730. The archive preserves both matched study-A sessions with full notes, operations and assessments. Neither the single trajectory nor its matched outcome establishes a self-confirmation cycle.

Source-only forecast references

Each reference fits only the session’s source final assays. Same-solvent matching uses exact sets of committed solvent IDs and falls back to the overall mean. It does not match quantities or thermal history. Five-world-equal errors and session wins are distinct summaries.

all: 42/45 valid sessions

Response	Agent	Source mean	Same solvent	Wins vs mean
Recovery	0.101292	0.204430	0.201849	39/42
Purity	0.038535	0.005099	0.006810	0/42
Size index	0.076867	0.023611	0.023732	5/42
Fines	0.319726	0.307554	0.290161	23/42

Fallback queries: 186/504.

Opaque: 15/15 valid sessions

Response	Agent	Source mean	Same solvent	Wins vs mean
Recovery	0.112728	0.201029	0.180902	13/15
Purity	0.016242	0.004742	0.006001	0/15
Size index	0.062510	0.019649	0.018896	1/15
Fines	0.322664	0.306509	0.280942	8/15

Fallback queries: 1/180.

Aligned: 14/15 valid sessions

Response	Agent	Source mean	Same solvent	Wins vs mean
Recovery	0.102667	0.195220	0.201526	13/14
Purity	0.051329	0.005525	0.006997	0/14
Size index	0.094661	0.032278	0.031455	2/14
Fines	0.309550	0.307566	0.301415	8/14

Fallback queries: 95/168.

MisIndexed: 13/15 valid sessions

Response	Agent	Source mean	Same solvent	Wins vs mean
Recovery	0.088987	0.219352	0.226733	13/13
Purity	0.052614	0.005112	0.007631	0/13
Size index	0.073908	0.018435	0.020434	2/13
Fines	0.327772	0.309612	0.290930	7/13

Fallback queries: 90/156.

Session-level source-reference comparisons

Session	Recovery: agent / mean	Purity: agent / mean	Size: agent / mean	Fines: agent / mean
01/O/1	0.022292 / 0.116932	0.014601 / 0.004324	0.084466 / 0.015104	0.473923 / 0.266733
01/A/1	- / -	- / -	- / -	- / -
01/M/1	0.124581 / 0.156698	0.012121 / 0.004324	0.068216 / 0.015104	0.415640 / 0.271286
02/O/1	0.045769 / 0.216456	0.022334 / 0.004468	0.035234 / 0.015674	0.372738 / 0.325952
02/A/1	0.111759 / 0.243664	0.017250 / 0.007786	0.046134 / 0.024788	0.169885 / 0.172576
02/M/1	0.149746 / 0.330325	0.008048 / 0.005389	0.073799 / 0.016439	0.355366 / 0.361589

Session	Recovery: agent / mean	Purity: agent / mean	Size: agent / mean	Fines: agent / mean
03/O/1	0.195531 / 0.235699	0.013328 / 0.007809	0.067725 / 0.042411	0.305874 / 0.368773
03/A/1	0.123152 / 0.091364	0.082770 / 0.007528	0.215642 / 0.162261	0.397820 / 0.354471
03/M/1	0.065694 / 0.137113	0.033687 / 0.004251	0.077693 / 0.021677	0.342820 / 0.354471
04/O/1	0.144967 / 0.206832	0.015497 / 0.004613	0.027652 / 0.017750	0.279968 / 0.345542
04/A/1	0.078039 / 0.182599	0.034210 / 0.005028	0.168319 / 0.015686	0.291282 / 0.261551
04/M/1	0.032967 / 0.163220	0.078627 / 0.004937	0.131236 / 0.015686	0.379615 / 0.276100
05/O/1	0.094099 / 0.252777	0.036949 / 0.004898	0.017929 / 0.020096	0.292398 / 0.349524
05/A/1	0.040578 / 0.202158	0.056949 / 0.004257	0.184953 / 0.019398	0.268231 / 0.329264
05/M/1	0.049862 / 0.174004	0.064449 / 0.005223	0.074659 / 0.019350	0.340731 / 0.322389
03/M/2	0.078031 / 0.336216	0.015796 / 0.004948	0.030984 / 0.021677	0.325304 / 0.354471
03/A/2	0.134469 / 0.135442	0.052404 / 0.004308	0.154776 / 0.023586	0.329070 / 0.354471
03/O/2	0.108864 / 0.218553	0.020796 / 0.004223	0.063526 / 0.021677	0.380304 / 0.354471
04/M/2	0.073123 / 0.158264	0.088627 / 0.005464	0.030819 / 0.015686	0.227948 / 0.261547
04/A/2	0.108295 / 0.155999	0.119460 / 0.007838	0.058893 / 0.070322	0.346388 / 0.409689
04/O/2	0.134807 / 0.185897	0.014460 / 0.004915	0.040237 / 0.015686	0.239615 / 0.252701
05/M/2	- / -	- / -	- / -	- / -
05/A/2	0.138265 / 0.227770	0.018616 / 0.004233	0.027319 / 0.019633	0.326565 / 0.329424
05/O/2	0.112432 / 0.108109	0.016594 / 0.005038	0.029829 / 0.019504	0.209065 / 0.322389
01/M/2	0.069362 / 0.282098	0.085450 / 0.005232	0.010132 / 0.024230	0.282257 / 0.471447
01/A/2	0.115339 / 0.236957	0.055450 / 0.006620	0.059466 / 0.020167	0.295189 / 0.379030
01/O/2	0.042862 / 0.266549	0.006213 / 0.004457	0.038130 / 0.021977	0.374807 / 0.411871
02/M/2	0.093995 / 0.227423	0.049815 / 0.008309	0.186033 / 0.015657	0.379533 / 0.179860
02/A/2	0.076759 / 0.158562	0.061482 / 0.006310	0.030086 / 0.015950	0.315399 / 0.172568
02/O/2	0.091829 / 0.196159	0.018565 / 0.004265	0.056715 / 0.015657	0.390399 / 0.170517
05/A/3	0.156345 / 0.234339	0.040282 / 0.004684	0.026100 / 0.020194	0.349065 / 0.348953
05/O/3	0.164099 / 0.263613	0.019449 / 0.004300	0.057159 / 0.019570	0.282389 / 0.329931
05/M/3	0.181599 / 0.310636	0.016077 / 0.004747	0.095370 / 0.020794	0.319898 / 0.358571
01/A/3	0.133330 / 0.192868	0.089617 / 0.005111	0.040880 / 0.015412	0.324406 / 0.339995
01/O/3	0.077293 / 0.179813	0.013302 / 0.005061	0.045299 / 0.015104	0.263090 / 0.240610
01/M/3	0.040496 / 0.114985	0.092117 / 0.004761	0.076966 / 0.017125	0.281022 / 0.221309
02/A/3	0.129746 / 0.256290	0.038149 / 0.004217	0.121866 / 0.021626	0.263572 / 0.172575
02/O/3	0.196412 / 0.286884	0.015380 / 0.004345	0.181866 / 0.015657	0.395238 / 0.245791
02/M/3	0.080208 / 0.205242	0.040649 / 0.004217	0.067688 / 0.015657	0.324533 / 0.247360
03/A/3	0.034869 / 0.202875	0.007131 / 0.005021	0.014927 / 0.021677	0.339471 / 0.354471
03/O/3	0.145116 / 0.180109	0.007203 / 0.004257	0.133980 / 0.023181	0.306685 / 0.360130
03/M/3	0.126364 / 0.290994	0.079853 / 0.004696	0.018982 / 0.021677	0.307820 / 0.354471
04/A/3	0.034721 / 0.192497	0.023627 / 0.004074	0.220377 / 0.015686	0.317115 / 0.274940
04/O/3	0.114550 / 0.101050	0.008962 / 0.004164	0.057902 / 0.015686	0.273462 / 0.252701
04/M/3	- / -	- / -	- / -	- / -

Separate research-opportunity comparison

Thirty sessions, 180/180 source batches, 117/120 assessment stages and 29 valid forecasts. The world-3 third Research session lost Q and later stages after a network stream failure. All source records and K1 are retained. No replacement was made. This is a total research-package effect, not a notebook ablation.

Initial scientific information

All thirty sessions use Opaque information. The exact public inputs contain the task and goals, response definitions, material catalogue, legal operations and instruments, resource limits and initial public state. The instance archival prior is null: no additional fitted local law or world-specific dossier is supplied. This does not remove public chemical framing or the model's pretrained knowledge. The archive includes all thirty full initial contexts. Fifteen within-world/repeat comparisons verify equality after removing only the campaign identifier. Research then receives twelve source batches and the notebook opportunity; NoResearch receives no source observations. Assessment prompts, order and restrictions match.

Session-level macro errors

World	Repeat	Opportunity	Batches	All MAE	Dilute MAE	Other MAE
01	1	Research	12	0.056735	0.203531	0.007803
01	1	NoResearch	0	0.097477	0.047073	0.114278
02	1	NoResearch	0	0.086437	0.007813	0.112645
02	1	Research	12	0.062682	0.162872	0.029285
03	1	Research	12	0.044852	0.151336	0.009358
03	1	NoResearch	0	0.097565	0.023106	0.122385
04	1	NoResearch	0	0.314567	0.279467	0.326267
04	1	Research	12	0.035504	0.107984	0.011344
05	1	Research	12	0.009812	0.015225	0.008008
05	1	NoResearch	0	0.110002	0.057774	0.127411
03	2	NoResearch	0	0.094482	0.011265	0.122222
03	2	Research	12	0.021299	0.011932	0.024421
04	2	Research	12	0.041674	0.148329	0.006122
04	2	NoResearch	0	0.107223	0.026615	0.134092
05	2	NoResearch	0	0.104525	0.030523	0.129193
05	2	Research	12	0.037601	0.134996	0.005136
01	2	NoResearch	0	0.104395	0.047076	0.123501
01	2	Research	12	0.059241	0.208909	0.009352
02	2	Research	12	0.011292	0.014620	0.010182
02	2	NoResearch	0	0.098404	0.028613	0.121667
05	3	Research	12	0.010708	0.015587	0.009082
05	3	NoResearch	0	0.089775	0.030412	0.109563
01	3	Research	12	0.055311	0.200086	0.007053
01	3	NoResearch	0	0.321575	0.278683	0.335872
02	3	NoResearch	0	0.118238	0.028397	0.148185
02	3	Research	12	0.047534	0.083833	0.035434
03	3	Research	12	-	-	-
03	3	NoResearch	0	0.116623	0.011366	0.151708
04	3	NoResearch	0	0.119995	0.013826	0.155385
04	3	Research	12	0.007154	0.014283	0.004778

Query group	Research	NoResearch	Difference
all	0.035632	0.132086	-0.096454
dilute	0.103677	0.061467	0.042210
other	0.012950	0.155625	-0.142675

Response-specific point errors

Within each world, valid sessions receive equal weight; worlds then receive equal weight. The macro error averages the three response errors. Differences are Research minus NoResearch, so positive values favour NoResearch. All/dilute/other contain twelve/three/nine queries. These are descriptive repeated-session results, not estimates from independent world samples.

Group	Response	Research	NoResearch	Difference
all	Normalized pH	0.012591	0.072952	-0.060362
all	Dissociation	0.057211	0.099741	-0.042530
all	Precipitation	0.037093	0.223563	-0.186470
dilute	Normalized pH	0.037733	0.029967	0.007766
dilute	Dissociation	0.191034	0.083088	0.107945
dilute	Precipitation	0.082265	0.071347	0.010918
other	Normalized pH	0.004210	0.087281	-0.083071
other	Dissociation	0.012604	0.105292	-0.092688
other	Precipitation	0.022036	0.274302	-0.252266

Dissociation accounts for 85.2% of the dilute macro-MAE increase, calculated as its response difference divided by the sum of the three response differences. This decomposition is not a claim that every session or query deteriorates. Interval outcomes below also distinguish point prediction from uncertainty quality.

Within-world response means: all queries

World	Arm	Valid n/3	Normalized pH	Dissociation	Precipitation
01	Research	3	0.021696	0.107957	0.041635
01	NoResearch	3	0.095485	0.172090	0.255872
02	Research	3	0.014875	0.059068	0.047564
02	NoResearch	3	0.040861	0.057124	0.205094
03	Research	2	0.010111	0.043408	0.045707
03	NoResearch	3	0.050835	0.050893	0.206941
04	Research	3	0.009800	0.044898	0.029634
04	NoResearch	3	0.106492	0.155320	0.279974
05	Research	3	0.006472	0.030725	0.020924
05	NoResearch	3	0.071089	0.063278	0.169935

Within-world response means: dilute queries

World	Arm	Valid n/3	Normalized pH	Dissociation	Precipitation
01	Research	3	0.072206	0.393844	0.146476
01	NoResearch	3	0.060239	0.144033	0.168560
02	Research	3	0.035797	0.164376	0.061151
02	NoResearch	3	0.003046	0.054014	0.007764
03	Research	2	0.030830	0.151906	0.062167
03	NoResearch	3	0.006663	0.033469	0.005606
04	Research	3	0.030088	0.151011	0.089497
04	NoResearch	3	0.060244	0.090590	0.169075
05	Research	3	0.019743	0.094031	0.052034
05	NoResearch	3	0.019643	0.093336	0.005730

Within-world response means: other queries

World	Arm	Valid n/3	Normalized pH	Dissociation	Precipitation
01	Research	3	0.004859	0.012661	0.006688
01	NoResearch	3	0.107234	0.181442	0.284975
02	Research	3	0.007901	0.023965	0.043035
02	NoResearch	3	0.053466	0.058160	0.270871
03	Research	2	0.003205	0.007243	0.040221
03	NoResearch	3	0.065559	0.056702	0.274053
04	Research	3	0.003037	0.009527	0.009679
04	NoResearch	3	0.121907	0.176897	0.316940
05	Research	3	0.002048	0.009623	0.010554
05	NoResearch	3	0.088237	0.053259	0.224671

Every planned session: all response errors

World	Arm	Repeat	Normalized pH	Dissociation	Precipitation
01	Research	1	0.021528	0.106668	0.042010
01	Research	2	0.023017	0.110089	0.044618
01	Research	3	0.020542	0.107113	0.038278
01	NoResearch	1	0.031158	0.080013	0.181260
01	NoResearch	2	0.031158	0.080180	0.201845
01	NoResearch	3	0.224139	0.356075	0.384510
02	Research	1	0.017729	0.083821	0.086495
02	Research	2	0.003079	0.014424	0.016371
02	Research	3	0.023817	0.078958	0.039826
02	NoResearch	1	0.046364	0.042754	0.170193
02	NoResearch	2	0.038110	0.064392	0.192709
02	NoResearch	3	0.038110	0.064225	0.252380
03	Research	1	0.016998	0.074242	0.043317
03	Research	2	0.003225	0.012575	0.048097
03	Research	3	-	-	-
03	NoResearch	1	0.056418	0.052566	0.183710
03	NoResearch	2	0.048002	0.050152	0.185294
03	NoResearch	3	0.048085	0.049963	0.251820
04	Research	1	0.011171	0.052384	0.042958
04	Research	2	0.015261	0.068802	0.040958
04	Research	3	0.002968	0.013510	0.004985
04	NoResearch	1	0.193608	0.366587	0.383507
04	NoResearch	2	0.066892	0.051399	0.203378
04	NoResearch	3	0.058975	0.047974	0.253036
05	Research	1	0.003044	0.015969	0.010423
05	Research	2	0.013661	0.059819	0.039323
05	Research	3	0.002711	0.016386	0.013027
05	NoResearch	1	0.077950	0.073449	0.178607
05	NoResearch	2	0.065783	0.057693	0.190100
05	NoResearch	3	0.069533	0.058693	0.141100

Every planned session: dilute response errors

World	Arm	Repeat	Normalized pH	Dissociation	Precipitation
01	Research	1	0.071539	0.390411	0.148643
01	Research	2	0.075539	0.400711	0.150476
01	Research	3	0.069539	0.390411	0.140310
01	NoResearch	1	0.005128	0.129744	0.006348
01	NoResearch	2	0.005128	0.129744	0.006357
01	NoResearch	3	0.170461	0.172611	0.492976
02	Research	1	0.051725	0.301043	0.135847
02	Research	2	0.004059	0.033376	0.006424
02	Research	3	0.051608	0.158710	0.041180
02	NoResearch	1	0.007059	0.009290	0.007091
02	NoResearch	2	0.001039	0.076376	0.008424
02	NoResearch	3	0.001039	0.076376	0.007776
03	Research	1	0.057663	0.276739	0.119607
03	Research	2	0.003997	0.027072	0.004727
03	Research	3	-	-	-
03	NoResearch	1	0.010997	0.052928	0.005393
03	NoResearch	2	0.004330	0.024072	0.005393
03	NoResearch	3	0.004663	0.023406	0.006030
04	Research	1	0.031399	0.175478	0.117076
04	Research	2	0.053799	0.246411	0.144776

World	Arm	Repeat	Normalized pH	Dissociation	Precipitation
04	Research	3	0.005066	0.031145	0.006639
04	NoResearch	1	0.154601	0.192724	0.491076
04	NoResearch	2	0.015399	0.056189	0.008257
04	NoResearch	3	0.010733	0.022855	0.007891
05	Research	1	0.005754	0.035553	0.004368
05	Research	2	0.048388	0.209653	0.146947
05	Research	3	0.005088	0.036887	0.004786
05	NoResearch	1	0.024088	0.143447	0.005786
05	NoResearch	2	0.017421	0.068780	0.005368
05	NoResearch	3	0.017421	0.067780	0.006035

The review archive also supplies every session's other-query results, coverage, interval width and interval score. Its 1,044 query-response rows retain the point estimate, stated 80% interval and withheld reference mean for all 29 valid forecasts. Nine empty session/group/response cells belong to the single missing forecast; they are neither zero errors nor additional failed sessions.

Interval outcomes

Nominal 80% intervals; equal-world means. Lower interval score is better. Coverage alone rewards broad intervals.

Group	Response	Arm	Coverage	Width	Interval score
all	Normalized pH	Research	0.918333	0.056723	0.077467
all	Dissociation	Research	0.900556	0.172358	0.338802
all	Precipitation	Research	0.782778	0.095436	0.202354
dilute	Normalized pH	Research	0.722222	0.128060	0.210358
dilute	Dissociation	Research	0.655556	0.440189	1.104098
dilute	Precipitation	Research	0.524444	0.131811	0.421346
other	Normalized pH	Research	0.983704	0.032944	0.033170
other	Dissociation	Research	0.982222	0.083081	0.083703
other	Precipitation	Research	0.868889	0.083311	0.129356
all	Normalized pH	NoResearch	0.983333	0.353500	0.355774
all	Dissociation	NoResearch	0.901111	0.471515	0.500389
all	Precipitation	NoResearch	0.915556	0.620167	0.727575
dilute	Normalized pH	NoResearch	1.000000	0.370889	0.370889
dilute	Dissociation	NoResearch	1.000000	0.872733	0.872733
dilute	Precipitation	NoResearch	0.933333	0.357111	0.417828
other	Normalized pH	NoResearch	0.977778	0.347704	0.350736
other	Dissociation	NoResearch	0.868148	0.337775	0.376274
other	Precipitation	NoResearch	0.909630	0.707852	0.830824

Missing-forecast sensitivity

Posthoc conservative [0,1] macro-MAE bounds for missing forecasts; fixed 3 sessions per arm/world, then 5 equal worlds. Not imputation or sampling uncertainty.

Group	Difference lower	Difference upper
all	-0.098659	-0.031992
dilute	0.036768	0.103434
other	-0.143801	-0.077134

The failed initial execution attempt is retained separately: four launched sessions, two completed source histories totalling 24 batches and no K1. An assessment-transport defect was fixed before the entire block restarted. Final-block recorded usage is 13,428,656 input-plus-output tokens, with 29/30 complete ledgers and one partial record; unavailable cost is not zero.

Retained matched-quench dispositions

Ninety planned pairs, 75 complete, all 180 trajectories replayed. Eighteen donor recipes are historical and not corrected-block participants. Effects are present minus absent, conditional on both final assays. Failed arms are not repaired. No agent receives the control outcomes.

Donor	Source world	Information	Repeat
1	01	Aligned	01
2	01	Opaque	01
3	03	Aligned	01
4	03	Opaque	01
5	05	Aligned	01
6	05	Opaque	01
7	01	Aligned	02
8	01	Opaque	02
9	03	Aligned	02
10	03	Opaque	02
11	05	Aligned	02
12	05	Opaque	02
13	01	Aligned	03
14	01	Opaque	03
15	03	Aligned	03
16	03	Opaque	03
17	05	Aligned	03
18	05	Opaque	03

Donor	World 1	World 2	World 3	World 4	World 5
1	Absent fails	Absent fails	Absent fails	Absent fails	Absent fails
2	0.000000	0.000000	0.000000	0.000000	0.000000
3	-0.071933	0.081214	-0.066411	-0.031221	-0.110806
4	0.304827	0.250542	0.268204	0.290060	0.000164
5	0.360502	0.751956	0.744909	0.702271	0.742834
6	0.000000	0.000000	0.000000	0.000000	0.000000
7	0.000000	-0.001465	0.000000	0.000000	0.000000
8	0.736430	0.676139	0.653463	0.615961	0.625517
9	Present fails	Present fails	Present fails	Present fails	Present fails
10	Present fails	Present fails	Present fails	Present fails	Present fails
11	0.177509	0.526305	0.525578	0.499879	0.576374
12	0.218968	0.147053	0.146991	0.075429	0.060371
13	0.005577	-0.050757	-0.092365	-0.084775	-0.218822
14	0.717709	0.644063	0.685473	0.628726	0.648716
15	0.602261	0.000216	0.617332	0.552729	0.347936
16	0.541322	0.424093	0.569429	0.546157	0.590242
17	0.573320	0.370738	0.548793	0.553698	0.495009
18	0.432735	0.470529	0.507058	0.528147	0.569570

Explanation histories and supporting contrasts

A bounded explanation-history panel

This retrospective panel follows the focal opportunity-study equilibrium session and four previously illustrated corrected-crystallization sessions. Seven related judgments are retained across five sessions, without claiming independent discoveries or population frequencies. The equilibrium session is not a member of the current 90-session window cohort. Full versions, exact quotations, action cursors and the unexecuted proposals accompany the portable evidence archive. Terminal revisions are included irrespective of whether research continues.

Judgment	First appearance	Observed test or limit	Subsequent standing and use
Weak-acid loading expectation	Before source experiments	Seven reagent batches, 0.05–0.8 M	Expected trend unsupported locally; alternative proposed
Component-loading explanation	After six assays	Five component-labelled batches, 0.1–0.8 M; same feed species	Intended chemical-identity contrast absent; local response remains flat
Background/weak-coupling account	Same post-six-assay note	Local plateau; broader discrimination unexecuted	Terminal mechanism unresolved; background favoured in dilute forecast
Lower-charge expectation	Before batch 12	Fines 0.529 to 0.387	Local recipe supported; batch 12 recommended
Volume interpolation	Before batch 12	Fines 0.512 to 0.601	Expectation retracted; earlier feasible batch recommended
Seed case B	Full version history retained	High-dose requests rejected	Explicitly recognizes unexecuted high-dose test
Seed case C	Full version history retained	High-dose requests rejected	Treats requested dose as tested negative evidence

The equilibrium command comparison was checked against the source version used for execution and the trajectory's mechanism identity. The mechanism has one positive non-catalytic feed; the reagent operation excludes catalyst and charges that feed, while component 0 selects the same feed directly. Both update the initial-loading ledger used by the equilibrium calculation. Equal-dose command changes therefore do not create a second acid identity. This limits discrimination between the proposed causes but does not erase evidence for the near-constant local response. The agent's terminal account retains fixed-background, saturation/clamping and below-resolution loading dependence as alternatives. Later use is checked separately against the sealed forecast rationale.

The lower-charge and interpolation examples are empirical expectations, not automatically explanatory-hypothesis generation. Their final notes matter: one retains a useful recipe while acknowledging catalyst/loading confounding; the other withdraws the interpolation and changes the recommendation. Seed records supply an execution-boundary contrast, rather than a third positive mechanism discovery. All interpretations are single-analyst retrospective readings, without an independent reliability estimate.

Corrected-crystallization distributions

The current 42 valid-forecast sessions supply 504 source final assays and 504 queries. Source purity has mean 0.986443, median 0.985187 and interquartile range 0.979441–0.994683; target entries have mean 0.984628 and range 0.974345–0.999518. Predictions average 0.947403, range 0.820000–0.995000, with 463/504 below reference means. The pooled signed error is −0.037226; averaging sessions within worlds and then the five worlds equally gives −0.037537. These are different aggregation conventions, both explicitly retained. Target entries repeat across sessions sharing a world. No independent-query inference or historical-cohort substitution is made. All four response distributions, signed errors, per-query forecasts and source-only reference calculations are provided; the three missing forecasts stay in the planned denominator of 45.

Research objectives and prior knowledge organize inquiry

Research starts from goals, available information and a constrained action space. The historical electrochemical and reaction studies ask agents either to develop and test scientific understanding or to optimize a public operating score. These commissions lead to genuine experimental achievements, but operating and predictive gains are not interchangeable (Fig. 2a,b). Optimization produces the better retested electrochemical procedure in 26 of 30 matched comparisons; among those 26, prediction error is lower in thirteen and higher in thirteen. In reaction processing, discovery produces both the better procedure and more accurate score predictions in seventeen of thirty pairs. The relation between a research objective and its scientific consequences therefore depends on what the inquiry actually establishes.

One electrochemical pair makes the distinction concrete (Fig. 2c,d). Discovery compares materials and conditions before increasing duration and current cap; optimization screens material combinations and then refines operating conditions for a promising pair. The optimization recommendation retests at 0.756 against discovery's 0.718, but its prediction mean absolute error (MAE) is 0.229 against 0.088. Its later prediction rationale describes several poorly performing cross-pair combinations as low-activity. That account accompanies a forecast of 0.070 against a reference of 0.494 for one combination under new conditions, while another forecast remains accurate at 0.570 against 0.572. The histories expose what a single endpoint obscures: success in developing a procedure can coexist with uneven judgments about the broader experimental space.

Prior information provides another entry into the organization of inquiry. In corrected crystallization, all fifteen aligned-information sessions start with solvent S2 and catalyst C1, whereas all fifteen sessions without an instance dossier start with S0 and C0. Under the specified mismatched dossier, thirteen start with S2 and fourteen with C1. Mean committed solvent coverage is 2.93, 4.00 and 2.40 for Aligned, Opaque and MisIndexed, respectively. These descriptive differences locate how supplied information enters material selection; wider coverage alone does not establish better learning. In reaction processing, partial parameter knowledge has a clear predictive benefit: Aligned reduces macro MAE from 0.102 to 0.057 under discovery and from 0.139 to 0.084 under optimization, improving on Opaque in four of five worlds under each commission.

The equilibrium prior comparison contrasts with the positive reaction results. In the original Sol campaigns, correct partial information lowers macro MAE on nine queries from 0.01008 to 0.00465, yet increases it on the three most dilute queries from 0.01833 to 0.15862. Both directions occur in all five worlds. A separate repeated cohort shows the same aggregate directions with different magnitudes: 0.00907 to 0.00512 on nine queries and 0.09880 to 0.13205 on the dilute subset. These cohorts are reported separately. The model comparison further limits the interpretation (Extended Data Fig. 1c): Luna and Terra improve with Aligned information in both regions, GPT-5.5 changes little, and Astra retains low dilute error across all three arms. The effect of partial knowledge is therefore conditional on both the researcher and the condition being judged; a local benefit does not determine its value elsewhere.

Matched-world investigations and explanatory histories

Thirty additional investigations cover five matched parameter pairs, two networks and three fresh sessions per world. All complete twelve source batches and three sealed stages. These sessions are separate from the historical 300, current process 90 and research-opportunity 30. Shared keyed observations are not independent noise replications. Neither physical separation nor an accurate forecast constitutes unique mechanism identification.

Quantity	Observed/planned
Full research and assessment chains	30/30
Final assays	360/360
Sealed account/forecast/retrospective stages	90/90
Query-response predictions	1,080/1,080
Notebook versions	80
Source replays	30/30
Own/partner-world replay trajectories	60/60
Assays separating >=1 / >=2 channels by 3 SD	326/360; 158/360
Sessions with >=2-channel separating assay	30/30

Physical information and prediction by world

Errors average twelve queries within sessions and three sessions within each world. The fraction is the proportion of actual final assays with at least two channels separated by three declared assay standard deviations under fixed-action paired-world replay. It is neither a significance test nor an agent success rate.

Pair/network	Separating fraction	Normalized pH	Free dissociation	Precipitation
P01-A	0.556	0.00191	0.00309	0.00253
P01-B	0.444	0.00248	0.00572	0.00267
P02-A	0.472	0.00270	0.00430	0.00387
P02-B	0.500	0.00196	0.00248	0.00164
P03-A	0.556	0.00314	0.00289	0.00272
P03-B	0.444	0.00134	0.00358	0.00147
P04-A	0.472	0.00122	0.00195	0.00250
P04-B	0.472	0.00185	0.00307	0.00173
P05-A	0.278	0.00187	0.00210	0.00394
P05-B	0.194	0.00112	0.00199	0.00123

Forecast intervals

The following average worlds equally. High coverage must be interpreted with widths; these values do not establish nominal calibration. Reference observations repeat across the three sessions within a world.

Response	MAE	80% coverage	Mean width
Normalized pH	0.00196	0.99333	0.02167
Free dissociation	0.00312	0.96000	0.01986
Precipitation	0.00243	0.98056	0.02096

Selected process anchors

Identifiers below locate evidence, not independent explanation counts. All thirty sessions were read; the selection illustrates distinct transitions rather than estimating their frequency. Public histories include all notes and sealed payloads.

Session	Transition	Temporal anchors
Repeat 3 / pair 3A	Threshold versus smooth relation; prospective tests; scoped forecast	Notes 0,36,72; subsequent batches 7-12; Q
Repeat 2 / pair 1A	Prospective dilution distinguishes concentration from persistence	Note 50; actions 51-56; final note 61; Q09
Repeat 1 / pair 2B	Rejected solvent creates apparent anomaly; terminal explanation	Rejection 11; notes 30 and terminal; K1/Q/K2
Repeat 3 / pair 4A	Incorrect concentration match followed by explicit withdrawal	Failures 9-11; note 25; final note 64; K1/Q08/K2

The additional retrospective reading covers 105 existing sessions and 309 notes, 112 selected focal histories and 218 verified note quotations. These 105 are the current ninety plus fifteen research-opportunity sessions, not new experiments. Four selected sealed follow-ups add thirteen verified anchors. The new thirty sessions contain 67 focal histories and 365 verified anchors. Both readings include terminal revisions, use a single analysis workflow and do not claim independent rater agreement or an abduction prevalence estimate.

Failures and reproducibility

Source attempts: 1,901 = 1,893 committed + six rolled back + two validation failures. The 131 recovered tool failures are separately counted. The first launch has four startup failures, twenty-six unstarted sessions and zero scientific operations; the entire fixed block restarts after repair. Earlier physical-candidate and canary failures remain in the source record. No completed scientific outcome is replaced. The review archive supplies public histories, all 1,080 predictions, 360 separation rows, per-world/per-session quantitative summaries, semantic readings and resource accounting. Existing exact replay is preserved; compilation runs no new experiment.

Historical broad-study tables

These 34 tables preserve the original numerical cells and denominators. They do not describe the corrected crystallization or opportunity cohorts. Historical crystallization recovery diagnostics retain their original protocol limitations. The original broad study comprises 300 campaigns; the five-model equilibrium subset overlaps that count. Earlier figure labels in table captions refer to the historical report, not current preview figures.

Retained table 1: A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	**Six system families**		**240**	**3,597 / 3,600**

Table A1. Cross-system study scope (GPT-5.6 Sol, medium). Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have scientific account (K1), sealed prediction (Q), and retrospective review (K2) stages; 238 meet the full source-and-assessment specification.

Retained table 2: A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

Tables 3-18 report MAE / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions remain included. PA intervals are nominally 90%; all others are 80%. Response scales and reference targets differ across systems.

Retained table 3: EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

Retained table 4: EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

Retained table 5: PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

Retained table 6: EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

Retained table 7: EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

Retained table 8: PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

Retained table 9: RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

Retained table 10: RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

Retained table 11: RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

Retained table 12: RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

Retained table 13: EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

Retained table 14: EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

Retained table 15: EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

Retained table 16: C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

Retained table 17: C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

Retained table 18: P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

Retained table 19: C.1 Equilibrium: a reversal across public concentration regimes

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Retained table 20: C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

Retained table 21: C.4 Original equilibrium campaigns: predictions and research chronology

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

Retained table 22: E.3 Complete crystallization baseline and interval comparisons

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response's original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the

public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Retained table 23: E.3 Complete crystallization baseline and interval comparisons

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign's twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

Retained table 24: E.4 Complete research-objective comparison

Table E3. All joint goal outcomes.

Higher retest	Lower score MAE	EC pairs	RX pairs
Optimization	Optimization	13	4
Optimization	Discovery	13	5
Discovery	Optimization	1	4
Discovery	Discovery	3	17

Both endpoints favour the same campaign in 16/30 EC and 21/30 RX pairs, and opposite campaigns in 14/30 and 9/30. Strict signs define these descriptive counts; near-zero differences are not thereby significant. Each system reuses five worlds. RX score MAE favours optimization in 8/30 pairs; its separate six-response macro MAE does so in 6/30.

Retained table 25: E.5 Strata and recovery sensitivity

Table E4. Endpoint orderings within design strata.

System	Factor	Level	Pairs	Same	Opposite
EC	World	W01	6	3	3
EC	World	W02	6	4	2
EC	World	W03	6	3	3
EC	World	W04	6	5	1
EC	World	W05	6	1	5
EC	Budget	12	15	7	8
EC	Budget	24	15	9	6
EC	Arm	Aligned	10	6	4
EC	Arm	MisIndexed	10	5	5
EC	Arm	Opaque	10	5	5
EC	Locus	E	30	16	14
RX	World	W01	6	4	2
RX	World	W02	6	4	2
RX	World	W03	6	3	3
RX	World	W04	6	4	2
RX	World	W05	6	6	0

System	Factor	Level	Pairs	Same	Opposite
RX	Budget	12	30	21	9
RX	Arm	Aligned	10	6	4
RX	Arm	MisIndexed	10	9	1
RX	Arm	Opaque	10	6	4
RX	Locus	P	15	12	3
RX	Locus	S	15	9	6

Same means that the higher-retest campaign also has lower MAE; opposite means it has higher MAE. These overlapping strata do not add independent tests. Budget rows compare goals at a fixed budget, not a twelve-to-twenty-four-batch contrast. E denotes entity priors, P parameter priors and S structural priors.

Retained table 26: E.6 All batches of the selected electrochemical pair

Table E5. Discovery source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	-1.5	100	1200	0.000000
2	S1/E0	-1.5	100	1200	0.167530
3	S2/E0	-1.5	100	1200	0.000000
4	S3/E0	-1.5	100	1200	0.050903
5	S0/E0	0.5	100	1200	0.509008
6	S0/E0	1.5	100	1200	0.044293
7	S0/E0	2.5	100	1200	0.030644
8	S0/E1	0.5	100	1200	0.000000
9	S0/E2	0.5	100	1200	0.482517
10	S0/E3	0.5	100	1200	0.360679
11	S0/E2	0.5	100	1200+13200	0.606942
12	S0/E2	0.5	500	1200+13200	0.710658

Retained table 27: E.6 All batches of the selected electrochemical pair

Table E6. Optimization source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	1	100	3600	0.535704
2	S1/E1	-1.5	300	7200	0.282124
3	S2/E2	1.5	300	7200	0.674932
4	S3/E3	1.5	300	7200	0.313717
5	S2/E0	1.5	300	7200	0.033808
6	S0/E2	1.5	300	7200	0.042443
7	S2/E2	1	300	7200	0.570485
8	S2/E2	2	300	7200	0.000000
9	S2/E2	1.5	500	7200	0.722737
10	S2/E2	1.5	500	10000	0.731956
11	S2/E2	1.4	500	9000	0.714786
12	S2/E2	1.5	400	12000	0.749194

Both campaigns select batch 12. Independent retests are 0.717587 for discovery and 0.756122 for optimization; twelve-query score MAEs are 0.088394 and 0.229397.

Retained table 28: E.8 Complete optimization forecasts

Table E7. Query settings, sealed forecasts and reference scores.

Q	Pair	V	mA	s	Forecast	80% interval	Reference
01	S0/E0	-0.8	100	600	0.050	0.000-0.180	0.032659
02	S0/E0	0.8	100	600	0.230	0.040-0.480	0.349491
03	S0/E0	-0.8	100	7200	0.250	0.080-0.440	0.358985
04	S0/E0	0.8	100	7200	0.570	0.430-0.680	0.572012
05	S0/E1	-0.8	100	7200	0.080	0.000-0.250	0.362359
06	S0/E1	0.8	100	7200	0.100	0.000-0.290	0.627599
07	S0/E3	-0.8	100	7200	0.070	0.000-0.230	0.347853
08	S0/E3	0.8	100	7200	0.090	0.000-0.280	0.542955
09	S2/E0	-0.8	100	7200	0.060	0.000-0.210	0.288226
10	S2/E0	0.8	100	7200	0.070	0.000-0.240	0.494376
11	S0/E0	-0.8	500	7200	0.170	0.040-0.360	0.276451
12	S0/E0	0.8	500	7200	0.550	0.390-0.680	0.344881

Score MAE is 0.229397; interval coverage is 5/12. The accurate Q04 prediction is retained alongside the large underestimates at Q06, Q08 and Q10. Source-to-query changes involve amount, volume and electrical settings, so these contrasts do not isolate one changed variable. Twelve query outcomes do not establish population calibration. No intermediate sealed prediction checkpoints were collected, so final prediction errors cannot be plotted as a within-session learning curve.

Retained table 29: F.1 Original intervals and campaign-specific observations

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 mL (nominal concentration 13.3 μM). The displayed source ranges are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the plateau. Public explanation categories appear in retained table 21 above.

Retained table 30: F.2 Sol interval and regime comparisons

Supplementary Table F2. GPT-5.6 Sol equilibrium predictions across test regions.

Arm	Remaining nine: MAE	Lowest three: MAE	Remaining nine: coverage	Lowest three: coverage
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

Values average five worlds and three response targets per group: normalized pH, dissociation fraction and precipitation signal. Coverage uses original 80% prediction intervals, with 675 reference-containment judgments per arm for the remaining nine conditions and 225 for the three lowest-concentration conditions; these judgments are not independent world replicates. The concentration grouping is post hoc and exploratory; the remaining nine conditions also include boundaries.

Retained table 31: F.3 Complete five-model comparison

Supplementary Table F3. Equilibrium predictions from five medium model configurations.

Model	Arm	Remaining nine: MAE	Lowest three: MAE	Lowest three: coverage
GPT-5.5	Opaque	0.00647	0.16325	6.7%
GPT-5.5	Aligned	0.00638	0.16146	2.7%
GPT-5.5	MisIndexed	0.00461	0.13604	13.3%
GPT-5.6 Luna	Opaque	0.09867	0.15240	48.4%
GPT-5.6 Luna	Aligned	0.04997	0.13319	31.1%
GPT-5.6 Luna	MisIndexed	0.05494	0.12556	44.4%
GPT-5.6 Terra	Opaque	0.03268	0.14611	67.1%
GPT-5.6 Terra	Aligned	0.01545	0.11466	26.2%
GPT-5.6 Terra	MisIndexed	0.01369	0.12130	41.8%
GPT-5.6 Sol	Opaque	0.01008	0.01833	93.3%
GPT-5.6 Sol	Aligned	0.00465	0.15862	2.2%
GPT-5.6 Sol	MisIndexed	0.00439	0.14397	13.8%
GPT-6 Astra	Opaque	0.00244	0.04040	82.7%
GPT-6 Astra	Aligned	0.00384	0.03853	82.7%
GPT-6 Astra	MisIndexed	0.00234	0.01857	92.4%

Each row contains five independent source campaigns, equally weighting the five worlds; worlds recur across information conditions and models. Each campaign predicts pH/14, dissociation fraction and precipitation signal for twelve queries. The three lowest-concentration conditions are the same fixed query group; the remaining nine also include boundary and unexplored conditions. Coverage concerns original 80% intervals and five reference observations, not confidence intervals or independent model replications. All three arms are retained; lower MisIndexed error does not itself establish a causal benefit of incorrect knowledge.

Retained table 32: G.1 Equilibrium reversal by response

Table G1. Original prediction MAE by response and concentration group.

Response	Group	Opaque	Aligned	MisIndexed	Lower / higher
pH / 14	Other nine	0.00261	0.00269	0.00254	2 / 3 of 5
pH / 14	Dilute three	0.00448	0.05576	0.05230	0 / 5 of 5
Dissociation fraction	Other nine	0.00839	0.00709	0.00660	3 / 2 of 5
Dissociation fraction	Dilute three	0.02640	0.28229	0.23861	0 / 5 of 5
Precipitation signal	Other nine	0.01923	0.00417	0.00403	5 / 0 of 5
Precipitation signal	Dilute three	0.02411	0.13783	0.14099	0 / 5 of 5

The final column counts worlds with lower/higher Aligned MAE than Opaque. Signed response differences divided by three sum to the macro-error difference. They quantify the contribution on the declared normalized scales; they do not establish a shared failure mechanism or physical comparability of targets.

Retained table 33: G.2 Separating purity offset from variation error

Table G2. Purity offset and variation diagnostics.

For each campaign, prediction error decomposes as MSE $= \overline{p-y}^{\,2} + \frac{1}{12}\sum_i[(p_i-\bar{p})-(y_i-\bar{y})]^2$. Centred RMSE is the square root of the second term. Demeaning uses withheld references for diagnosis, not as an available predictor.

Batches	Arm	N	Bias	cRMSE	Ref. SD	Pred. SD	Offset share
Both	All	30	-4.289	1.889	0.598	1.968	87.6%
12	Opaque	5	-0.806	0.970	0.598	1.031	45.3%
12	Aligned	5	-4.920	1.837	0.598	1.791	89.8%
12	MisIndexed	5	-9.166	2.843	0.598	2.977	91.3%
24	Opaque	5	-1.816	1.682	0.598	1.649	57.4%
24	Aligned	5	-3.948	1.797	0.598	1.983	85.6%
24	MisIndexed	5	-5.080	2.206	0.598	2.376	85.3%

N is the campaign count; cRMSE denotes centred RMSE. Bias, cRMSE and both standard deviations are campaign means in percentage points. Offset share is the ratio of summed squared campaign biases to summed campaign MSEs, with equal query counts; it is not the mean campaign fraction. The median campaign fraction is 77.6%. Mean bias is negative in 30 of 30 campaigns. Centred RMSE exceeds reference SD in 29 of 30. These comparisons describe forecast outputs and do not identify an internal revision process. No new agent or simulator calls were made.

Retained table 34: H.2 A concrete three-arm information example

Table H1. Public prior content in one matched physical world.

Supplied field	Opaque	Aligned	MisIndexed
Effective pKa interval (80%)	Absent	[4.6094, 4.7094]	[5.3294, 5.4294]
Dilution change in pH/14	Absent	+0.003494987	+0.000875889
Dilution change in dissociation	Absent	+0.008187416	+0.001884676
Nominal confidence	Absent	0.8	0.8

The common dilution anchor is 0.001 mol acid at 298.15 K, with volume increasing from 0.018 to 0.054 L. Both supplied records identify the source as an "independent bounded archival fit" and qualify the claim as a "local effective relationship; not a universal aqueous-chemistry law". The pKa intervals above are rounded for display; the input contains their full precision. Aligned therefore supplies correct partial local information, not the evaluator's complete process equations, query answers or an instruction to enter the most dilute test region. The experiment contrasts the resulting complete research processes, including possible changes to evidence acquisition and interpretation.

Executed process relationships

The following equations describe the retained virtual-world laws, not mechanisms inferred by the participants. The initial-charge final-assay recovery formula below also describes the checked full-process observation route in the repeated cohort. The internal stage-recovery metric and mismatched participant description are distinguished above; they must not be substituted for the observed/scored response.

Equilibrium response model

The parameter-prior study uses a bounded monoprotic acid-base model followed by a precipitation calculation. These are the process relations used for the retained campaigns and references. They are distinct from explanations proposed by the agents. In particular, an agent's proposed active-pool cap is not the implemented source of the dissociation plateau.

Let n be the current ledger amount of the initial reactant, V the current sample volume and $C = n/V$. The background-ion amounts are $n_+ = \min(0.020, 0.15n)$ and $n_- = \min(0.020, 0.08n)$, in moles. With $H = [\mathrm{H}^+]$, $K_a = 10^{-\mathrm{p}K_a}$ and activity-ratio factor g, the solver obtains

$$[\mathrm{A}^-] = \frac{CK_a}{K_a + gH}, \qquad [\mathrm{OH}^-] = \frac{K_w(T)}{H},$$

subject to charge balance,

$$H + \frac{n_+}{V} - \frac{K_w(T)}{H} - [\mathrm{A}^-] - \frac{n_-}{V} = 0.$$

The reported responses are $\mathrm{pH}/14 = -\log_{10}(H)/14$ and dissociation fraction $\alpha = [\mathrm{A}^-]/C$, clipped to the public unit interval. In the uncapped background-ion range, charge balance implies $\alpha = 0.07 + H/C - K_w/(HC)$. At sufficiently

high concentration the last two terms are small, giving a local plateau near 7%; dilution makes their contribution appreciable. The plateau and departure therefore arise under the same fixed equations. The five effective pKa values span 4.6594 to 5.3794 in increments of 0.18; the default activity-ratio factor is one.

Precipitation is calculated sequentially from effective ion inventories $a = n_{+} + 0.10 n_{\mathrm{A}^-}$ and $b = n_{-} + 0.08 n_{\mathrm{HA}}$. When supersaturated, a 1:1 removal x, bounded by the limiting inventory, satisfies $(a - x)(b - x)/V^2 = K_{sp}$, with default $K_{sp} = 1.8 \times 10^{-10}$. The public signal is x/n, clipped to $[0, 1]$. This hook does not feed precipitation back into the acid-base solution. Thus the three public responses are related but should not be interpreted as three independent measurements of one failure mechanism.

Nominal concentration in the retained source comparisons is a recipe descriptor: total nominal reagent additions divided by total solvent additions. The solver instead uses the ledger state at measurement, after any sample consumption and other operations. The scatterplot combines distinct recipes and histories; it is not a controlled one-dimensional titration curve. The three most dilute test recipes have nominal concentrations of approximately 1.33×10^{-4}, 1.33×10^{-5} and 1.67×10^{-4} M. No source assay in these fifteen Sol campaigns reaches those concentrations.

Crystallization response model

The retained crystallization block composes upstream reaction, thermal operations, seeding, crystal nucleation and growth, filtration and measurement. It uses the latent-material process family and its linear supersaturation-dependent impurity transfer. Solid product, solid impurity and the particle population persist across operations; dissolved and solid inventories are accounted for separately.

The temperature-dependent solubility follows

$$c^*(T) = c^*_{\mathrm{ref}} \exp\left[-\frac{\Delta H}{R}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right],$$

with $T_{\mathrm{ref}} = 298.15$ K, $\Delta H = 20{,}000$ J mol^{-1} and a declared temperature domain of 250-430 K. World and solvent properties scale the reference solubility and kinetics. For relative supersaturation $s = \max(C/c^* - 1, 0)$, primary nucleation scales as $B = k_b s^2$ and linear growth as $G = k_g s$. Before world and solvent multipliers, the coefficients are 2×10^7 L^{-1} s^{-1} for nucleation and 2×10^{-8} m s^{-1} for growth. Nucleation and growth consume available dissolved target and update the population distribution; they are not independent score adjustments.

Each crystallized target increment transfers impurity according to

$$\Delta n_{I,s} = \min(n_{I,l},\ \Delta n_{P,s}\, k_{\mathrm{occ}}(1 + 0.5s)),$$

where $k_{\mathrm{occ}} = 0.02$ times the solvent's occlusion multiplier. The four nominal solvent multipliers are 1.20, 0.80, 0.65 and 1.05; world-fixed material residuals further modulate these properties. Filtration retains 96% of target solids and 92% of impurity solids. The final-assay purity is the solid-target mole fraction $n_{P,s}/(n_{P,s} + n_{I,s})$. Seed contributes to target purity but is subtracted in the full-process recovery numerator; final-assay recovery is normalized by the initial reagent charge.

These relations allow thermal and seeding histories to alter the amount and size distribution of crystals while impurity incorporation remains modest. They explain how recovery and particle responses can vary without requiring a comparable purity change. The approximately 98.5% purity level is an empirical property of the retained source and query conditions, not a hard-coded value or a universal invariant of the model. The present analysis does not isolate individual process contributions to that stability. The retained diagnostics test whether the forecasts preserve the level and variation of the retained purity references.